A counterexample to Oertel's conjecture on Mixed-integer volume

The sharp finite-dimensional form

Abstract

We construct a rational polytope $C \subset \mathbb{R}^{12} \times \mathbb{R}^5$ of positive mixed-integer volume and a number $r < (5/6)^5/2^{12}$ such that every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a proper closed halfspace whose intersection with C has at most the fraction r of that volume. Of the 4852 nonempty integer fibers of C, 4095 are heavy, one per nonzero parity class, and every point of a heavy fiber lies in a tilted cut that meets no other integer fiber. Three coordinated cuts cover the origin fiber, and each has positive mass only on that fiber and on one of three pools of 252 light fibers. The volume comparisons use exact rational arithmetic, and Lean 4 verifies the uniform depth bound.

1 The result and the construction

Let vol denote Lebesgue measure. The *mixed-integer volume*, or *mass*, of a set $D \subseteq \mathbb{R}^n \times \mathbb{R}^d$ with measurable slices is $\mu(D) = \sum_{\mathbf{z} \in \mathbb{Z}^n} \mathrm{vol}\{\mathbf{x} \in \mathbb{R}^d : (\mathbf{z}, \mathbf{x}) \in D\}$, and the *depth* of a mixed-integer point in a convex body C of positive mass is the infimum of $\mu(C \cap H)/\mu(C)$ over proper closed halfspaces H containing it. Conjecture 17 of Basu and Oertel [1] asserts that every convex body in $\mathbb{R}^n \times \mathbb{R}^d$ of positive mass has a point in $\mathbb{Z}^n \times \mathbb{R}^d$ of depth at least $2^{-n}(d/(d+1))^d$. We disprove this bound at $(n, d) = (12, 5)$, where it equals $(5/6)^5/2^{12}$. Cheng and Basu [2] use the name Oertel's conjecture in their Conjecture 2.3 for the lower bound $1/(e2^n)$, which we do not address.

Theorem 1.1. There exist a rational polytope $C \subset \mathbb{R}^{12} \times \mathbb{R}^5$ with nonempty interior, $\mathbf{0} \in C$ and $0 < \mu(C) < \infty$, and a real number r with $0 < r < (5/6)^5/2^{12}$, such that the following holds. For every $(\mathbf{z}, \mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5$ there are $b \in \mathbb{R}$ and a nonzero linear functional F on $\mathbb{R}^{12} \times \mathbb{R}^5$ with $F(\mathbf{z}, \mathbf{x}) \geq b$ and $\mu(C \cap \{F \geq b\}) \leq r\,\mu(C)$.

The proof compares a common budget $M > 0$ with the total mass $W = \mu(C)$. If every mixed-integer point lies in a proper closed halfspace, or *cut*, of mass at most M, then $r = M/W$ works, and $r < (5/6)^5/2^{12}$ is equivalent to $W > 4096M/(5/6)^5$. We write $M = (5h/6)^5$, so that the target is $W > 4096h^5$. Every point of a five-dimensional simplex has a barycentric coordinate at least $1/6$, and the cap where that coordinate is at least $1/6$ is a homothetic copy with ratio $5/6$. In a simplex of volume h^5 this cap has mass exactly M. Thus 4096 such simplices, each cap meeting only its own simplex, would only reproduce the conjectured fraction, so the construction must carry more mass while no cut exceeds M.

The polytope C is the convex hull of fibers over 4852 integer points, which we call bases. One heavy base lies in each of the 4095 nonzero classes of $(\mathbb{Z}/2\mathbb{Z})^{12}$, with a fiber of volume at most h^5. Each is a vertex of the convex hull of the bases, so every point of its fiber lies in a cap that can be tilted in the integer directions to miss all other fibers. The zero class contains the origin, whose fiber has volume $1 < h^5$ in our units. To make up the missing mass we add 756 light bases, each the negative of a heavy base, so that the two have the origin as their midpoint. Because of this, a tilt in the integer coordinates alone cannot separate the origin from both, so three coordinated cuts cover the origin fiber instead. Two selector coordinates sort the light bases into three pools of 252, and each cut has positive mass only on the origin fiber and on one pool. All three pools count toward W while each cut contains only one, and their total volume exceeds the amount by which the origin fiber and the heavy fibers with light partners fall short of h^5. To keep the origin fiber

unchanged in the convex hull, we place each fiber over $\mathbf{z}$ inside the dilation of the origin fiber by $1 + g(\mathbf{z})$ for a linear form g, so that antipodal bases have radii $1 + g$ and $1 - g$.

Section 2 defines the bases and the three linear forms that offset the cuts. Section 3 derives the fibers from the requirements of these cuts. Section 4 proves that C has no other integer fibers, and Section 5 constructs the cut at every point and completes the exact comparison. Appendix A records the Lean statement and the replay.

2 Integer bases and offset forms

Write $\mathbf{z} = (\mathbf{y}, \mathbf{s}) \in \mathbb{R}^{10} \times \mathbb{R}^2$, with ordinary coordinates $\mathbf{y}$ and selector coordinates $\mathbf{s}$, and put $u = \sum_i y_i$. The linear forms

$$\tau(\mathbf{z}) = (u - s_2,\ u - s_1,\ u - s_1 - s_2)$$

will offset the three cuts over the origin, and the sum of their values, $3u - 2s_1 - 2s_2$, will control the radii of the fibers.

Two choices make τ useful. Let $\mathbf{e}_1, \mathbf{e}_2$ be the unit vectors of $\mathbb{R}^2$. The selectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2$ have the three nonzero residues of $(\mathbb{Z}/2\mathbb{Z})^2$, and each will carry a separate copy of the ten-dimensional Boolean cube. Let $c_i = -1$ for $i \le 4$ and $c_i = 1$ for $i \ge 5$. For $J \subseteq \{1, \dots, 10\}$, let $\mathbf{1}_J$ be the indicator vector of J, and let $\mathbf{b}_J$ have entries c_i on J and 0 off J. The sign change does not alter the residue modulo 2, but it moves the ordinary sum. Since $u(\mathbf{b}_J) = |J \triangle \{1, 2, 3, 4\}| - 4$, toggling membership in the first four positions shows that $u = k$ has $\binom{10}{k+4}$ solutions. In particular, $u = 1$ has $\binom{10}{5} = 252$ solutions, whereas the unsigned masks $\mathbf{1}_J$ have only 10. The *candidates* are $(\mathbf{1}_J, \mathbf{0})$ for $J \neq \emptyset$ and $(\mathbf{b}_J, \mathbf{s})$ for $\mathbf{s} \in \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$, the latter with the following values.

Type j	$\mathbf{s}$	$\tau(\mathbf{b}_J, \mathbf{s})$	Sum of entries	Paired	Near
1	$\mathbf{e}_1$	$(u,\ u-1,\ u-1)$	$3u - 2$	$u = 1$	$u = 0$
2	$\mathbf{e}_2$	$(u-1,\ u,\ u-1)$	$3u - 2$	$u = 1$	$u = 0$
3	$\mathbf{e}_1 + \mathbf{e}_2$	$(u-1,\ u-1,\ u-2)$	$3u - 4$	$u = 1$	$u = 2$

The sum of the entries of τ is not divisible by 3 in this table and equals $3|J|$ at $(\mathbf{1}_J, \mathbf{0})$, so it is nonzero on every candidate. A *heavy base* is a candidate multiplied by the sign of this sum. After this orientation, the paired value of u in type j gives $\tau_j = 1$ and zeros elsewhere, the near value gives $\tau_j = 0$ and ones elsewhere, and every other value of u gives three positive entries, as do the bases $(\mathbf{1}_J, \mathbf{0})$. We call these heavy bases *paired* of type j, *near* of type j and *regular*, respectively. For each paired base $\mathbf{z}$ of type j, the point $-\mathbf{z}$ is a *light* base of type j, with $\tau_j = -1$ and zeros elsewhere. The set $\mathcal{Z}$ of *bases* consists of $\mathbf{0}$ and all heavy and light bases.

The orientation is easiest to see on one mask. For $J = \{1, 5, 6\}$, the vector $\mathbf{b}_J$ has entries $-1, 1, 1$ in positions $1, 5, 6$, and $u = 1$. Then $\tau(\mathbf{b}_J, \mathbf{e}_1) = (1, 0, 0)$ and $\tau(\mathbf{b}_J, \mathbf{e}_2) = (0, 1, 0)$, so these two candidates are paired bases as they stand. However, $\tau(\mathbf{b}_J, \mathbf{e}_1 + \mathbf{e}_2) = (0, 0, -1)$ has negative sum, so the paired base of type 3 is $-(\mathbf{b}_J, \mathbf{e}_1 + \mathbf{e}_2)$ with $\tau = (0, 0, 1)$, and its light partner sits at the positive selector. In the fourth parity class of this mask, $(\mathbf{1}_J, \mathbf{0})$ is a regular base with $\tau = (3, 3, 3)$. Figure 1(b) shows these seven bases and, for reference, the origin.

Sorting all bases by selector gives Figure 1(a). For types 1 and 2 the sum is positive exactly when $u \ge 1$, so the heavy base lies at selector $\mathbf{s}$ for $u \ge 1$ and at $-\mathbf{s}$ for $u \le 0$. For type 3 the sum is positive exactly when $u \ge 2$, so the heavy base lies at $\mathbf{e}_1 + \mathbf{e}_2$ for $u \ge 2$ and at $-(\mathbf{e}_1 + \mathbf{e}_2)$ for $u \le 1$. Summing $\binom{10}{u+4}$ gives $252 + 386$ paired and regular bases at each of $\mathbf{e}_1, \mathbf{e}_2, -(\mathbf{e}_1 + \mathbf{e}_2)$, and $210 + 176$ near and regular bases at each of their antipodes, which also hold 252 light bases.

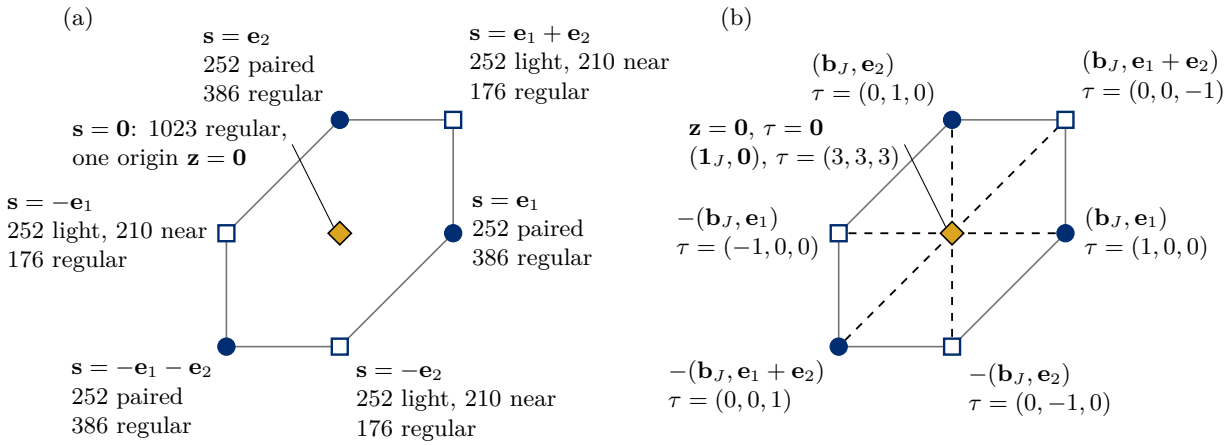


Figure 1: The image of $\mathcal{Z}$ under the selector projection $(\mathbf{y}, \mathbf{s}) \mapsto \mathbf{s}$. Each marked point is a layer of bases, and only these seven selectors occur, with the hexagon as their convex hull. Circles mark the three layers with paired bases, squares the antipodal layers with light bases, and the diamond the layer $\mathbf{s} = \mathbf{0}$. (a) The census of each layer, which gives $3 \cdot 638 + 3 \cdot 638 + 1024 = 4852$. The diamond aggregates 1024 fibers, those of the 1023 regular bases $(\mathbf{1}_J, \mathbf{0})$ and of the unique origin $\mathbf{z} = \mathbf{0}$. (b) The seven bases of the mask $J = \{1, 5, 6\}$, with the origin as a reference. Labels give the full bases, including the sign of the ordinary part. The paired bases of types 1 and 2 lie at their positive selectors, whereas the paired base of type 3 lies at its negative selector $-(\mathbf{e}_1 + \mathbf{e}_2)$ and has ordinary part $-\mathbf{b}_J$. A dashed segment joins antipodal bases, whose midpoint is the full origin $\mathbf{z} = \mathbf{0}$. The base $(\mathbf{1}_J, \mathbf{0})$ has the same selector as the origin but is a different point of $\mathbb{Z}^{12}$.

Modulo 2, an oriented $(\mathbf{b}_J, \mathbf{s})$ reduces to $(\mathbf{1}_J, \mathbf{s})$, and the three selectors have distinct nonzero residues. Hence the heavy bases represent each nonzero class of $(\mathbb{Z}/2\mathbb{Z})^{12}$ exactly once, and there are 4095 of them, namely 756 paired, 630 near and $3 \cdot 386 + 3 \cdot 176 + 1023 = 2709$ regular. A light base $-\mathbf{z}$ has its partner's residue and differs from it, so the 4852 points of $\mathcal{Z}$ are distinct.

3 The simplex, the radii and the fibers

Let $B = \{\mathbf{x} \in \mathbb{R}^6 : x_i \geq 0,\ \sum_i x_i = 1\}$ be the standard simplex with vertices $\mathbf{v}_1, \ldots, \mathbf{v}_6$. We identify the hyperplane $\sum_i x_i = 1$ with $\mathbb{R}^5$ by dropping x_6 and subtracting $1/6$ from the other coordinates. The centroid of B is then the origin of $\mathbb{R}^5$, scalar multiples of subsets are dilations about the centroid, and $x_1, \ldots, x_6$ are affine functions on $\mathbb{R}^5$ with sum 1. From now on vol and μ are measured in units of the Lebesgue volume of B, which does not change the ratios in Theorem 1.1.

We introduce the parameters of the construction with the conditions they must satisfy. Their exact values are fixed in Section 5, and until then only these conditions are used. Fix a rational number a with $1/2 < a < 1$. For $j = 1, 2, 3$, with these three indices taken cyclically, put $f_j = x_j - a x_{j+1} - (1 - a) x_{j+2}$. Each f_j vanishes at the centroid and equals 1 at $\mathbf{v}_j$, so it is a nonzero linear functional on $\mathbb{R}^5$. Moreover $|f_j| \leq 1$ on B and $f_1 + f_2 + f_3 = 0$. The cyclic permutation of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is a volume-preserving linear symmetry of B that carries f_j to f_{j+1}, so we describe the sets below for $j = 1$ and obtain the others by rotation. The three cuts over the origin are the proper closed halfspaces

$$H_j = \{(\mathbf{z}, \mathbf{x}) \in \mathbb{R}^{12} \times \mathbb{R}^5 : f_j(\mathbf{x}) \geq 3\tau_j(\mathbf{z})\}, \qquad j = 1, 2, 3.$$

Over $\mathbf{z} = \mathbf{0}$ their traces are $B \cap \{f_j \geq 0\}$, and these cover B because $f_1 + f_2 + f_3 = 0$. At another

base, the value of τ_j decides what the fiber may look like. If $\tau_j(\mathbf{z}) \geq 1$, then H_j misses a fiber on which $f_j < 3$. If $\tau_j(\mathbf{z}) = 0$, then the trace of H_j on a fiber contained in $\{f_j \leq 0\}$ lies in the hyperplane $\{f_j = 0\}$, which is null. If $\tau_j(\mathbf{z}) = -1$, then H_j contains a fiber on which $f_j \geq -3$. So, by Section 2, a paired or light fiber of type j should lie in $\{f_i \leq 0\}$ for both $i \neq j$, a near fiber of type j should lie in $\{f_j \leq 0\}$, and a regular fiber needs no sign condition. These conditions pass through dilations about the origin because the f_j are linear.

The convex hull of the fibers must also leave the fiber over the origin unchanged. Fix a rational number δ with $0 < \delta < 1$ and put $g(\mathbf{z}) = \delta(\tau_1(\mathbf{z}) + \tau_2(\mathbf{z}) + \tau_3(\mathbf{z})) = \delta(3u - 2s_1 - 2s_2)$. We will place each fiber over $\mathbf{z}$ inside $(1 + g(\mathbf{z}))B$. Since g is linear, antipodal bases get the radii $1 + g$ and $1 - g$, and Lemma 4.2 shows that the origin fiber stays B under arbitrary convex combinations. This is the design we use, and it is not forced for fibers of other shapes. Here g equals δ at paired bases, $-\delta$ at light bases, 2δ at near bases and at least 3δ at regular bases.

Each f_j is a homogeneous linear form in (x_1, x_2, x_3). Non-strict inequalities $f_j \geq 0$ or $f_j \leq 0$ therefore cut out a join of their trace on the active face $\operatorname{conv}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ with $\operatorname{conv}(\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$. Indeed, normalizing the first three coordinates by their sum when it is positive preserves the inequalities, and points with that sum zero belong to the latter triangle. The simplex spanned by three points of the active face and $\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ has volume equal to the absolute determinant of the 3×3 matrix of barycentric coordinates of these points, because the affine map carrying the vertices of B to the vertices of this simplex acts on barycentric coordinates as the direct sum of this matrix and the identity. We call this the determinant formula. On the active face, f_1 is positive at $\mathbf{v}_1$, negative at $\mathbf{v}_2$ and $\mathbf{v}_3$, and zero at $(a\mathbf{v}_1 + \mathbf{v}_2)/(1 + a)$ and $((1 - a)\mathbf{v}_1 + \mathbf{v}_3)/(2 - a)$. So $B \cap \{f_1 \geq 0\}$ is a simplex, and the determinant formula gives its volume $\beta = 1/[(1 + a)(2 - a)]$.

The sign conditions and the radius confine a paired or light fiber of type j to a dilation of $T_j = B \cap \{f_{j+1} \leq 0,\ f_{j+2} \leq 0\}$, on which $f_j = -f_{j+1} - f_{j+2} \geq 0$. Among the vertices of the active face, f_2 is positive only at $\mathbf{v}_2$ and f_3 only at $\mathbf{v}_3$. The lines $f_2 = 0$ and $f_3 = 0$ meet at the centroid $\bar{\mathbf{v}} = (\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3)/3$ of the face. They meet the edges from $\mathbf{v}_1$ to $\mathbf{v}_2$ and from $\mathbf{v}_1$ to $\mathbf{v}_3$ at $\mathbf{w}_2 = (\mathbf{v}_1 + (1 - a)\mathbf{v}_2)/(2 - a)$ and $\mathbf{w}_3 = (\mathbf{v}_1 + a\mathbf{v}_3)/(1 + a)$, respectively. The edge from $\mathbf{v}_2$ to $\mathbf{v}_3$ has $f_1 < 0$ throughout, so it is disjoint from T_1. The trace of T_1 is the quadrilateral with vertices $\mathbf{v}_1, \mathbf{w}_2, \bar{\mathbf{v}}, \mathbf{w}_3$ in Figure 2(a), whose diagonal from $\mathbf{v}_1$ to $\bar{\mathbf{v}}$ lies on the line $x_2 = x_3$. Consequently $T_1 = U \cup P$, where

$$U = \operatorname{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6), \qquad P = \operatorname{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_2, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6).$$

These simplices lie on opposite sides of the hyperplane $x_2 = x_3$ and meet in a common facet, which is a null set. By the determinant formula, U has volume $\theta = a/[3(1 + a)]$ and P has volume $\phi = (1 - a)/[3(2 - a)]$.

A light fiber can be all of $(1 - \delta)T_j$, because the only cut we use on it is H_j, whose mass is computed as a whole. A paired fiber needs caps of mass at most M. We start from the simplex $(1 + \delta)U$ because a simplex has an explicit cap bound. At our parameters its volume falls short of h^5, and the total deficit over the 756 paired bases exceeds the light mass. We therefore add a thin slab of P along the common facet, which restores most of the volume and keeps the cap calculation tractable. Section 5 gives the exact volumes.

To define the slab, let $\nu_0, \ldots, \nu_5$ be the barycentric coordinates of U in the listed vertex order and put $q_i = \nu_i(\mathbf{w}_2)$. Comparing active coordinates gives $q_1 = 3(1-a)/(2-a)$, $q_2 = -(1-a^2)/[a(2-a)]$, $q_0 = 1 - q_1 - q_2$ and $q_3 = q_4 = q_5 = 0$. Since $1/2 < a < 1$, we have $0 < q_1 < 1$, $q_2 < 0$ and $0 < q_1 + q_2 = q_1(2a - 1)/(3a) < 1$, so $0 < q_0 < 1$. Fix a rational number ε with $0 < \varepsilon < 1/1000$ and put $K_1 = T_1 \cap \{\nu_2 \geq \varepsilon q_2\}$. On U we have $\nu_2 \geq 0 > \varepsilon q_2$. On P, let p be the barycentric coordinate at $\mathbf{w}_2$ and, for $i \neq 2$, let p_i be the barycentric coordinate at the vertex of U with index i. Then

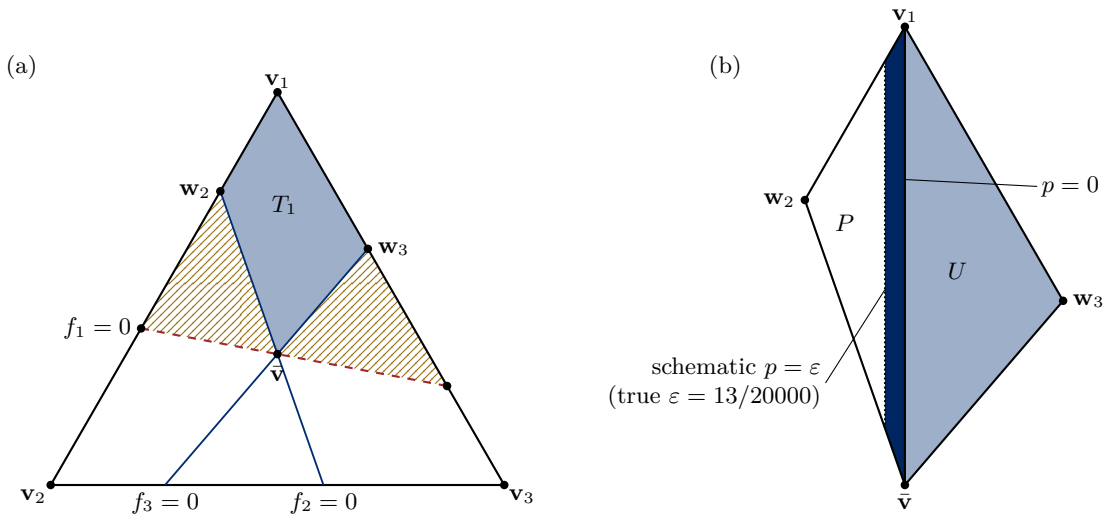


Figure 2: Traces on the active face $x_4 = x_5 = x_6 = 0$ at $a = 663/1000$. These are not pictures of five-dimensional volumes. The point $\bar{\mathbf{v}}$ is the centroid of this face, and the centroid of B, about which all fibers are dilated, lies off this face. (a) To scale, with points given in barycentric coordinates of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. The dashed line $f_1 = 0$ meets the edges at $(663, 1000, 0)/1663$ and $(337, 0, 1000)/1337$. The triangle above it is the trace of the cap $B \cap \{f_1 \geq 0\}$, of volume β, which is the origin part of H_1. Inside this triangle, the shaded quadrilateral is the trace of the smaller set T_1, of volume $\theta + \phi$, with $\mathbf{w}_2 = (1000, 337, 0)/1337$ and $\mathbf{w}_3 = (1000, 0, 663)/1663$, and hatching marks the rest of the cap. (b) The active-face trace of T_1 enlarged uniformly by the factor $7/4$. The diagonal $p = 0$ separates U from P, and K_1 retains U and the dark strip of P with $p \leq \varepsilon$. The strip width is exaggerated, drawn at $p = 1/5$ for the true $\varepsilon = 13/20000$.

$\nu_i = p_i + q_i p$ for $i \neq 2$ and $\nu_2 = q_2 p$, since both sides are affine and agree at the vertices of P. Therefore K_1 is the union of U and $P \cap \{p \leq \varepsilon\}$. The set $P \cap \{p \geq \varepsilon\}$ is a homothetic copy of P with ratio $1 - \varepsilon$, so $\mathrm{vol}(K_1) = V := \theta + \phi[1 - (1 - \varepsilon)^5]$. Rotation gives K_2, K_3 and their coordinates ν_i. The inequality $\nu_2 \geq \varepsilon q_2$ has a nonzero constant term in the homogeneous coordinates (x_1, x_2, x_3), so K_1, unlike T_1, U, P, is not a join of its active trace with $\mathrm{conv}(\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$. Figure 2(b) shows only that trace.

A near base of type j has radius $1 + 2\delta$ and needs a fiber in $\{f_j \leq 0\}$. Fix a rational number $h > 0$ with $2h < 1 + 2\delta$ and put $\eta = (h/(1+2\delta))^5$. Then $\eta < 1/32$, and $a > 1/2$ gives $a/(1+a) > 1/3$, so $\eta < a/(1+a)$. Let $N_j = \mathrm{conv}\big(\eta \mathbf{v}_j + (1-\eta)\mathbf{v}_{j+1}, \mathbf{v}_{j+1}, \mathbf{v}_{j+2}, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6\big) \subseteq B$. At these vertices f_j takes the values $\eta - a(1-\eta)$, $-a$, $a - 1$, 0, 0, 0, which are nonpositive because $\eta < a/(1+a)$. Hence $f_j \leq 0$ on N_j, and $\mathrm{vol}(N_j) = \eta$ by the determinant formula. We assign the following fibers to the bases.

Base $\mathbf{z}$	Number	$g(\mathbf{z})$	$\tau(\mathbf{z})$	Fiber $A_{\mathbf{z}}$	$\mathrm{vol}(A_{\mathbf{z}})$
$\mathbf{0}$	1	0	$\mathbf{0}$	B	1
paired, type j	252 per type	δ	$\tau_j = 1$, others 0	$(1+\delta)K_j$	$(1+\delta)^5 V$
light, type j	252 per type	$-\delta$	$\tau_j = -1$, others 0	$(1-\delta)T_j$	$(1-\delta)^5(\theta+\phi)$
near, type j	210 per type	2δ	$\tau_j = 0$, others 1	$(1+2\delta)N_j$	h^5
regular	2709	$\geq 3\delta$	all entries ≥ 1	hB	h^5

The volume of a near fiber is $(1 + 2\delta)^5 \eta = h^5$. Every fiber lies in $(1 + g(\mathbf{z}))B$, and the radius $1 + g(\mathbf{z})$ is positive because $\delta < 1$. For regular bases this containment uses $h < 1 + 2\delta < 1 + g(\mathbf{z})$

and $\mathbf{0} \in B$. Every heavy fiber lies in $(1+2\delta)B$, so $|f_j| \le 1+2\delta < 3$ on it because $\delta < 1$. Let $C = \operatorname{conv}\big(\bigcup_{\mathbf{z}\in\mathcal{Z}} \{\mathbf{z}\} \times A_{\mathbf{z}}\big)$. The light fibers of one type have total volume $L = 252(\theta+\phi)(1-\delta)^5$, and all fibers have total volume $W = 3339h^5 + 756(1+\delta)^5 V + 1 + 3L$. The comparison with the target of Section 1 therefore reads

$$W - 4096h^5 = (1-h^5) + 756\big[(1+\delta)^5 V - h^5\big] + 3L. \tag{1}$$

4 The integer fibers of C

We first show that a nonzero integer point of $\operatorname{conv}(\mathcal{Z})$ has only the trivial representation.

Lemma 4.1. If $\mathbf{z} \in \mathbb{Z}^{12} \setminus \{\mathbf{0}\}$ is a convex combination $\sum_m \lambda_m \mathbf{z}^m$ of points $\mathbf{z}^m \in \mathcal{Z}$ with all $\lambda_m > 0$, then $\mathbf{z}^m = \mathbf{z}$ for every m.

Proof. All entries of all bases lie in $\{-1, 0, 1\}$, so $\mathbf{z} = (\mathbf{y}, \mathbf{s})$ has the same property. If a coordinate of $\mathbf{z}$ equals ± 1, or if the contributors $\mathbf{z}^m = (\mathbf{y}^m, \mathbf{s}^m)$ take only two consecutive integer values in that coordinate, then integrality forces all contributors to agree in it. We call this saturation.

Suppose first that $\mathbf{s} \neq \mathbf{0}$, say $s_k = \pm 1$. Saturation gives $s_k^m = s_k$ for all m. Every nonzero selector of a base is $\pm\mathbf{e}_1$, $\pm\mathbf{e}_2$ or $\pm(\mathbf{e}_1 + \mathbf{e}_2)$, so the other selector coordinate of each contributor lies in $\{0, s_k\}$, and saturation gives $\mathbf{s}^m = \mathbf{s}$. Let $\omega = \pm 1$ be the sign of the nonzero entries of $\mathbf{s}$. By construction, a base with this selector has ordinary part $\omega \mathbf{b}_J$, so $y_i^m \in \{0, \omega c_i\}$ for each i, and saturation gives $\mathbf{y}^m = \mathbf{y}$.

Now suppose that $\mathbf{s} = \mathbf{0}$. Then $y_i = \pm 1$ for some i, and saturation gives $y_i^m = y_i$ for all m. A contributor with a nonzero selector of sign ω has $y_i = \omega c_i$, so $c_i y_i (s_1^m + s_2^m) = \omega(s_1^m + s_2^m) > 0$, while a contributor with selector $\mathbf{0}$ gives 0. The weighted sum of these numbers is $c_i y_i (s_1 + s_2) = 0$, so all contributors have selector $\mathbf{0}$. Their ordinary parts lie in $\{0,1\}^{10}$, and saturation gives $\mathbf{y}^m = \mathbf{y}$. □

Consequently $\mathbb{Z}^{12} \cap \operatorname{conv}(\mathcal{Z}) = \mathcal{Z}$, and no nonzero base lies in the convex hull of the other bases. For a point $\mathbf{d}$ outside a compact convex set, let $\mathbf{d}_0$ be a nearest point in the set. Minimizing the squared distance to $\mathbf{d}$ on segments from $\mathbf{d}_0$ gives $\langle \mathbf{d} - \mathbf{d}_0, \mathbf{d}' - \mathbf{d}_0 \rangle \le 0$ for every point $\mathbf{d}'$ of the set. The inner product with $\mathbf{d} - \mathbf{d}_0$ is thus strictly larger at $\mathbf{d}$ than on the set. In particular, each nonzero $\mathbf{z} \in \mathcal{Z}$ has a linear functional $\ell_{\mathbf{z}}$ on $\mathbb{R}^{12}$ with $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for all $\mathbf{z}' \in \mathcal{Z} \setminus \{\mathbf{z}\}$.

Lemma 4.2. The set C is a rational polytope with nonempty interior and $\mathbf{0} \in C$. For $\mathbf{z} \in \mathbb{Z}^{12}$, the slice $\{\mathbf{x} : (\mathbf{z}, \mathbf{x}) \in C\}$ equals $A_{\mathbf{z}}$ if $\mathbf{z} \in \mathcal{Z}$ and is empty otherwise. Consequently $\mu(C) = W$, and every closed halfspace H satisfies $\mu(C \cap H) = \sum_{\mathbf{z}\in\mathcal{Z}} \operatorname{vol}\big(A_{\mathbf{z}} \cap \{\mathbf{x} : (\mathbf{z}, \mathbf{x}) \in H\}\big)$.

Proof. The fibers are polytopes with rational vertices, because T_j and K_j are cut from B by rational inequalities, N_j has rational vertices, and the dilation factors are rational. So C is the convex hull of finitely many rational points. It contains $\{\mathbf{0}\} \times B$, hence the origin. The set B spans the continuous directions. The regular bases $(\mathbf{1}_{\{i\}}, \mathbf{0})$ and the near bases $(\mathbf{0}, -\mathbf{e}_1)$ and $(\mathbf{0}, -\mathbf{e}_2)$ span the integer directions, so C has nonempty interior.

Let $(\mathbf{z}, \mathbf{x}) \in C$ with $\mathbf{z}$ integer, and write $(\mathbf{z}, \mathbf{x}) = \sum_m \lambda_m (\mathbf{z}^m, \mathbf{x}^m)$ with $\lambda_m > 0$, $\sum_m \lambda_m = 1$, $\mathbf{z}^m \in \mathcal{Z}$ and $\mathbf{x}^m \in A_{\mathbf{z}^m}$. If $\mathbf{z} \neq \mathbf{0}$, then Lemma 4.1 gives $\mathbf{z}^m = \mathbf{z}$ for all m, so $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$ by convexity of $A_{\mathbf{z}}$. If $\mathbf{z} = \mathbf{0}$, then $\mathbf{x}^m/(1+g(\mathbf{z}^m)) \in B$ for all m. Since g is linear, the positive weights $\lambda_m(1+g(\mathbf{z}^m))$ sum to $1 + g(\mathbf{z}) = 1$. Thus $\mathbf{x}$ is a convex combination of the points $\mathbf{x}^m/(1+g(\mathbf{z}^m))$ of B with these weights, which gives $\mathbf{x} \in B = A_{\mathbf{0}}$. The reverse inclusions are immediate, and the formulas for μ follow by summing over $\mathcal{Z}$. □

5 Cuts and the exact comparison

We now construct the halfspaces, starting with the three cuts over the origin, which contain the light fibers.

Lemma 5.1. For $j = 1, 2, 3$ we have $\mu(C \cap H_j) = \beta + L$. Each point $(\mathbf{0}, \mathbf{x})$ with $\mathbf{x} \in B$ and each point of a light fiber lies in some H_j.

Proof. We read the trace of H_j on each fiber from the table of Section 3, using Lemma 4.2. Over the origin the trace is $B \cap \{f_j \geq 0\}$, of volume β. If $\tau_j(\mathbf{z}) = -1$, the condition $f_j \geq -3$ holds on the whole light fiber of type j, and these fibers have total volume L. If $\mathbf{z} \neq \mathbf{0}$ and $\tau_j(\mathbf{z}) = 0$, then $\mathbf{z}$ is a paired or light base of a type $k \neq j$, or a near base of type j. Its fiber lies in $\{f_j \leq 0\}$, so the trace lies in the null hyperplane $\{f_j = 0\}$. If $\tau_j(\mathbf{z}) \geq 1$, then $\mathbf{z}$ is heavy and the condition $f_j \geq 3$ fails on its fiber. Hence $\mu(C \cap H_j) = \beta + L$. Since $f_1 + f_2 + f_3 = 0$, each $\mathbf{x} \in B$ has $f_j(\mathbf{x}) \geq 0$ for some j, so $(\mathbf{0}, \mathbf{x}) \in H_j$. A point of a light fiber of type j lies in H_j. □

Tilting a cap in the integer directions handles the heavy bases.

Lemma 5.2. Let $\mathbf{z} \neq \mathbf{0}$ be a base, let ψ be a nonconstant affine function on $\mathbb{R}^5$, and let $\kappa \in \mathbb{R}$. There are a nonzero linear functional F and $b \in \mathbb{R}$ with $F(\mathbf{z}, \mathbf{x}') \geq b$ if and only if $\psi(\mathbf{x}') \geq \kappa$, and with $\mu(C \cap \{F \geq b\}) = \mathrm{vol}(A_{\mathbf{z}} \cap \{\psi \geq \kappa\})$.

Proof. For $\rho > 0$ put $F(\mathbf{z}', \mathbf{x}') = \psi(\mathbf{x}') - \psi(\mathbf{0}) + \rho \ell_{\mathbf{z}}(\mathbf{z}')$ and $b = \kappa - \psi(\mathbf{0}) + \rho \ell_{\mathbf{z}}(\mathbf{z})$. Then F is a nonzero linear functional, and $F(\mathbf{z}, \mathbf{x}') \geq b$ if and only if $\psi(\mathbf{x}') \geq \kappa$. Since $\mathcal{Z}$ is finite, the fibers are compact and $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for $\mathbf{z}' \neq \mathbf{z}$, we have $F < b$ on every fiber over $\mathbf{z}' \neq \mathbf{z}$ once ρ is large. Lemma 4.2 gives the mass. □

A regular or near fiber is a simplex of volume h^5. Some barycentric coordinate of a point in it is at least $1/6$. That coordinate is affine and nonconstant, and the cap of the simplex where it is at least $1/6$ is a homothetic copy with ratio $5/6$, so Lemma 5.2 gives a halfspace of mass $(5h/6)^5$ containing the point.

A paired fiber is not a simplex, and we cover it with the coordinates ν_i. Fix positive rational numbers $t_0, \ldots, t_5$ with $\sum_i t_i = 1$ and $1/7 < t_i < 1/5$. Let $\mathbf{z}$ be a paired base of type j and let $\mathbf{x} \in (1 + \delta)K_j$. By rotation we compute with $j = 1$. Since $\sum_i \nu_i = \sum_i t_i = 1$ on all of $\mathbb{R}^5$, there is an index i with $\nu_i(\mathbf{x}/(1 + \delta)) \geq t_i$. The function $\mathbf{x}' \mapsto \nu_i(\mathbf{x}'/(1 + \delta))$ is affine and nonconstant, so Lemma 5.2 with $\kappa = t_i$ gives a halfspace containing $(\mathbf{z}, \mathbf{x})$ of mass $(1+\delta)^5 V_i$, where $V_i = \mathrm{vol}(K_1 \cap \{\nu_i \geq t_i\})$.

Lemma 5.3. We have $V_2 = \theta(1 - t_2)^5$, and for $i \neq 2$,

$$V_i = \theta(1 - t_i)^5 + \phi \, \frac{(1 - t_i)^5 - \big(1 - t_i - (1 - q_i)\varepsilon\big)^5}{1 - q_i}.$$

Proof. We split K_1 into U and $P \cap \{p \leq \varepsilon\}$, which overlap in a null set. In U the cap $\{\nu_i \geq t_i\}$ is a homothetic copy of U with ratio $1 - t_i$, of volume $\theta(1 - t_i)^5$. On P with $i = 2$ we have $\nu_2 = q_2 p \leq 0 < t_2$, so this part is empty. Let $i \neq 2$. For $0 \leq p \leq \varepsilon$, the bounds $0 \leq q_i < 1$, $\varepsilon < 1/1000$ and $1/7 < t_i < 1/5$ give $t_i - q_i p > 0$, $1 - t_i - (1 - q_i)p > 0$ and $1 - q_i > 0$. The affine chart with coordinates p and p_k, $k \neq i, 2$, sends P onto the standard simplex of coordinate Lebesgue volume $1/120$, so volume in our units on P is 120ϕ times coordinate Lebesgue measure. Substituting $p_i = 1 - p - \sum_{k \neq i,2} p_k$ turns the condition $\nu_i = p_i + q_i p \geq t_i$ into $\sum_{k \neq i,2} p_k \leq 1 - t_i - (1 - q_i)p$, and it forces $p_i \geq t_i - q_i p > 0$. So for each p the four remaining coordinates range over a simplex

of coordinate Lebesgue volume $[1-t_i-(1-q_i)p]^4/24$. Multiplying by 120ϕ and integrating over p, the contribution of P is

$$5\phi\int_0^{\varepsilon}\big[1-t_i-(1-q_i)p\big]^4\,dp=\phi\,\frac{(1-t_i)^5-\big(1-t_i-(1-q_i)\varepsilon\big)^5}{1-q_i}.$$ □

The cuts above reduce the theorem to comparisons of explicit rational numbers. We take

$$a=\frac{663}{1000},\qquad \delta=\frac{664201}{10^6},\qquad \varepsilon=\frac{13}{20000},\qquad h=\frac{1111946415}{10^9},$$
$$(t_0,\dots,t_5)=10^{-9}(166735371,\ 166735294,\ 166324902,\ 166734811,\ 166734811,\ 166734811).$$

They satisfy all the conditions imposed so far, namely $1/2<a<1$, $0<\delta<1$, $0<\varepsilon<1/1000$, $h>0$, $2h<1+2\delta$, $\sum_i t_i=1$ and $1/7<t_i<1/5$. In lowest terms, $\theta=221/1663$, $\phi=337/4011$, $\beta=1000000/2223431$, $q_0=776569/886431$, $q_1=1011/1337$, $q_2=-560431/886431$ and $5h/6=74129761/80000000$. Exact evaluation of the formulas of Sections 3 and 5 gives the following floors.

Mass	$\lfloor 10^{12}\cdot\text{mass}\rfloor$	Margin	$\lfloor 10^{12}\cdot\text{margin}\rfloor$
h^5	1699884034250	$M-\beta-L$	1262439
M	683145268393	$M-(1+\delta)^5V_0$	7542
β	449755355574	$M-(1+\delta)^5V_1$	6288
L	233388650379	$M-(1+\delta)^5V_2$	1211
$3L$	700165951139	$M-(1+\delta)^5V_i,\ i=3,4,5$	7869
$(1+\delta)^5\theta$	1696402554056	$h^5-(1+\delta)^5V$	312774
$(1+\delta)^5V$	1699883721475	$1+3L-h^5$	281916889
W	6962725049747587	$W-4096h^5$	45459473

The left column shows what the slab of Section 3 does. The simplex $(1+\delta)U$ falls short of h^5 by more than $3\cdot10^{-3}$, so its total deficit over 756 bases exceeds $3L$, while $(1+\delta)K_j$ falls short by less than $4\cdot10^{-7}$. Each paired fiber thus remains slightly smaller than a regular one. In (1), the light mass $3L$ exceeds the sum of the two deficits h^5-1 and $756[h^5-(1+\delta)^5V]$, and the surplus is the last margin. The comparisons we need are $\beta+L<M$, $(1+\delta)^5V_i<M$ for all i, and $4096h^5<W$.

Proof of Theorem 1.1. Lemma 4.2 gives the stated properties of C and $\mu(C)=W$ in our units, and $0<W<\infty$ because W is a finite sum of positive volumes. A point $(\mathbf{z},\mathbf{x})\in\mathbb{Z}^{12}\times\mathbb{R}^5$ either lies outside C or has $\mathbf{z}\in\mathcal{Z}$ and $\mathbf{x}\in A_{\mathbf{z}}$. All inequalities defining the covers are nonstrict, so points on the boundaries of fibers need no separate treatment. If $(\mathbf{z},\mathbf{x})\notin C$, nearest-point separation gives a nonzero F with $F(\mathbf{z},\mathbf{x})>F$ on C. With $b=F(\mathbf{z},\mathbf{x})$, the halfspace $\{F\ge b\}$ contains the point and misses C, so it has mass 0. If $\mathbf{z}=\mathbf{0}$ or $\mathbf{z}$ is light, Lemma 5.1 gives a containing H_j of mass $\beta+L<M$. If $\mathbf{z}$ is regular or near, the simplex cap gives mass M. If $\mathbf{z}$ is paired, the cover by the ν_i gives mass $(1+\delta)^5V_i<M$. So every point of $\mathbb{Z}^{12}\times\mathbb{R}^5$ lies in a proper closed halfspace H with $\mu(C\cap H)\le M$.

Put $r=M/W=(5/6)^5h^5/W$. Then $\mu(C\cap H)\le r\,\mu(C)$, and $W>4096h^5$ gives $0<r<(5/6)^5/2^{12}$. Numerically, $10^5r=9.811464096\dots$ and $10^5(5/6)^5/2^{12}=9.811464160\dots$. □

A Lean statement and exact replay

The exact checker and Lean project are tracked in PR #14. The checker and Lean sources are unchanged since the verification source below, which remains the pin for them.

Verification source	`d368306f28ce808d2b2b773cc025745d835538bf`
Lean toolchain	`leanprover/lean4:v4.34.1`
mathlib revision	`d13f23b723b8a846827a245b89c10fc7d3f11612`

In the namespace `SharpOC.Selector17`, the final declaration reads as follows.

```
theorem sharp_oc_counterexample :
    ∃ C : Set Ambient, IsCompact C ∧ Convex ℝ C ∧ (interior C).Nonempty ∧
      ((0 : IntegerSpace), (0 : E)) ∈ C ∧
      0 < mixedIntegerMass C ∧ mixedIntegerMass C ≠ ⊤ ∧
      ∃ r : ℝ, 0 < r ∧ r < (((5 : ℝ) / 6)^5 / (2 : ℝ)^12) ∧
        ∀ (z : Coordinate → ℤ) (x : E),
          ∃ F : Ambient →L[ℝ] ℝ, ∃ b : ℝ, F ≠ 0 ∧ b ≤ F ((fun j => (z j : ℝ)), x) ∧
            (mixedIntegerMass (C ∩ {p | b ≤ F p})).toReal /
              (mixedIntegerMass C).toReal ≤ r
```

Here `Coordinate` is `Fin 10 ⊕ Fin 2`, `IntegerSpace` is `Coordinate → ℝ`, `E` is `Fin 5 → ℝ` and `Ambient` is `IntegerSpace × E`. For a set `D`, the value `mixedIntegerMass D` is the extended nonnegative sum, over all `w : Coordinate → ℤ`, of the Lebesgue volume of `{x | ((fun k => (w k : ℝ)), x) ∈ D}`, which is $\mu(D)$ in the Lebesgue units of Section 1. Finiteness of the mass makes both `toReal` values faithful. Lemma 4.2 establishes rationality of the polytope. With Python 3, Git and elan installed, run the following from the repository root. Lake reads its dependency pins from `lean/lake-manifest.json`.

```
set -eu
python3 -B research/sharp-dimension-17/check.py
cd lean
lake exe cache get
lake build
lake env lean CheckAxioms.lean
lake env lean --stdin <<'LEAN'
import SharpOC.Selector17Counterexample

#check SharpOC.Selector17.integer_projection_census
#check SharpOC.Selector17.ambient_dimension
#check SharpOC.Selector17.sharp_oc_counterexample
#print axioms SharpOC.Selector17.sharp_oc_counterexample
LEAN
```

The checker uses exact rational arithmetic to generate the 4852 listed bases and their counts. It also asserts the containment and cap inequalities together with $M-\beta-L > 10^{-6}$, $M-\max_i(1+\delta)^5 V_i > 10^{-9}$ and $W - 4096h^5 > 4 \cdot 10^{-5}$, which imply the three comparisons of Section 5. The floors in that section come from direct exact evaluation of the same formulas. The checker does not show that the list exhausts the integer projection. That statement is proved in Section 4 and in Lean by `integer_projection_census`. The Lean build passed, and this lemma, the dimension lemma and the final theorem depend only on `propext`, `Classical.choice` and `Quot.sound`.

References

[1] Amitabh Basu and Timm Oertel. *Centerpoints: A link between optimization and convex geometry*. Manuscript version with Conjecture 17.

[2] Hongyu Cheng and Amitabh Basu. *Linear Threshold for Oertel's Conjecture on the Mixed-Integer Volume*. Conjecture 2.3.